

CHAPTER SIX

RANDOM VARIABLE AND MATHEMATICAL EXPECTATION

Dr. Mohammad Manjur Alam (Manju)
Associate Professor
Department of Computer Science and Engineering
International Islamic University Chittagong

Random variable with example.

A random variable is a real valued function whose values are determined with the outcomes of a random experiment. It is usually denoted by X, Y, Z etc and the value of the random variable denoted by x, y, z .

Let us consider the experiment of tossing two fair coins. The sample space of the experiment is $S = \{HH, HT, TH, TT\}$

Let X denotes the number of heads. So,

Sample point	Number of Head (X)
HH	2
HT	1
TH	1
TT	0

Here, X can take the values 0, 1 and 2. Therefore X is a random variable.

Types of random variable.

There are two types of random variable:

1. Discrete random variable.
2. Continuous random variable.

Discrete random variable: A random variable is called discrete random variable if it can take only isolated values.

Example: Family members, mobile number etc.

Continuous random variable: A random variable is called continuous random variable if it can take any values between certain limits.

Example: Age, weight, height etc.

Probability function and probability density function.

Probability function: A function $f(x)$ of a discrete random variable X is called a probability function if it satisfies the following two conditions:

- (i) $f(x) \geq 0$
- (ii) $\sum f(x) = 1$

Probability density function: A function $f(x)$ of a continuous random variable X is called a probability **density** function if it satisfies the following two conditions:

- (i) $f(x) \geq 0$
- (ii) $\int_{-\infty}^{\infty} f(x) dx = 1$

Cumulative Distribution Function: The **cumulative distribution function (cdf)** $F(x)$ of a discrete random variable X with probability distribution $f(x)$ is $F(x) = P(X \leq x) = \sum_{t \leq x} f(t)$

Cumulative Distribution Function: The cumulative distribution function $F(x)$ of a continuous random variable X with density function $f(x)$ is $F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$ for $-\infty < x < \infty$

The consequence of above definition

$$P(a < x < b) = \int_{-\infty}^b f(x) dx - \int_{-\infty}^a f(x) dx = F(b) - F(a) \quad \text{and} \quad f(x) = \frac{dF(x)}{dx}$$

Application problem: A discrete random variable x has the following probability function:

x	0	1	2	3	4
f(x)	k	2k	3k	3k	k

For what value of k the function will be a probability function?

Ans: Since, $\sum f(x) = 1$

$$\text{Or, } (k + 2k + 3k + 3k + k) = 1$$

Or, $10k = 1$

Or, $k = 0.10$

Therefore,

x	0	1	2	3	4
f(x)	0.10	0.20	0.30	0.30	0.10

Application problem: A discrete random variable X has the following probability function:

Values of X:x	0	1	2	3	4
f(x)	0.12	0.18	k	0.30	0.16

(i) Find the value of k, Compute (ii) $P[X > 3]$; (iii) $P[1 < X < 4]$; and (iv) $P[X < 1]$.

(i) Since, $\sum f(x) = 1$

Or, $(0.76 + k) = 1$

Or, $k = 1 - 0.76$

Or, $k = 0.24$

(ii) $P[X > 3] = P[X = 4] = 0.16$

(iii) $P[1 < X < 4] = P[X = 2] + P[X = 3]$

$= k + 0.30$

$= 0.24 + 0.30 = 0.54$

(iv) $P[X < 1] = P[X = 0]$

$= 0.12$

Assignment problem: A discrete random variable X has the following probability function:

Values of X:x	0	1	2	3	4
f(x)	0.10	0.30	0.20	0.25	0.15

Find the value of (i) $P[X = 1]$; (ii) $P[X > 3]$; (iii) $P[1 < X < 4]$; and (iv) $P[X < 1]$.

Assignment problem: Suppose x is discrete random variable with probability function.

Values of X:x	-2	-1	0	1	2
f(x)	0.3	0.2	0.1	0.25	0.15

Find the value of (i) $P[X = 1]$; (ii) $P[-1 < X < 2]$; (iii) $P[X > 0]$ and $P[X < -1]$

Application problem: A continuous random variable X has the following probability density function:

$$f(x) = kx^2; 0 \leq x \leq 1$$

- (i) Find the value of K
- (ii) probability that X lies between 0.2 and 0.50
- (iii) probability that X less than 0.30 and
- (iv) probability that X greater than 0.75

Ans:(i) Since, $\int_{-\infty}^{\infty} f(x)dx = 1$

$$\text{Or, } \int_0^1 kx^2 dx = 1$$

$$\text{Or, } k \left[\frac{x^3}{3} \right]_0^1 = 1$$

$$\text{Or, } \frac{k}{3} = 1$$

$$\text{Or, } k = 3$$

- (ii) probability that X lies between 0.2 and 0.50

$$\begin{aligned} P[0.20 < X < 0.50] &= \int_{0.20}^{0.50} kx^2 dx \\ &= k \left[\frac{x^3}{3} \right]_{0.20}^{0.50} \\ &= 3 \left[\frac{1}{3} (0.50^3 - 0.20^3) \right] \\ &= 0.125 - 0.008 = 0.117 \end{aligned}$$

- (iii) probability that X less than 0.30

$$\begin{aligned} P[X < 0.3] &= \int_0^{0.3} kx^2 dx \\ &= k \left[\frac{x^3}{3} \right]_0^{0.30} = 3 \left[\frac{1}{3} (0.30^3 - 0^3) \right] = 0.027 \end{aligned}$$

$$\begin{aligned} \text{(iv) probability that } X \text{ greater than } 0.75 \text{ is } P[X > 0.75] &= \int_{0.75}^1 kx^2 dx \\ &= k \left[\frac{x^3}{3} \right]_{0.75}^1 = 3 \left[\frac{1}{3} (0.30^3 - 0^3) \right] = 0.578 \end{aligned}$$

Assignment problem: The following is the probability density function of a random variable x :

$$f(x) = \frac{3}{4}(2x - x^2); 0 < x < 2$$

Find (i) the value of 'K' ; (ii) $P(x > 1)$ and (iii) $P(1.5 < x < 2.25)$

Assignment problem: A continuous random variable X has the following probability density function:

$$f(x) = K(x - 1); \quad 2 \leq x \leq 6$$

Compute (i) the value of 'K' ; (ii) $P(X > 3)$ and (iii) $P(3 < X < 4)$

Assignment problem: A continuous random variable X has the following probability density function:

$$f(x) = K(x + 1); \quad 2 \leq x \leq 5$$

Compute (i) the value of 'K' ; (ii) $P(X > 3)$; (iii) $P(X = 4)$ and (iv) $P(3 < X < 4)$

Assignment problem: Let X be a continuous random variable with probability density function

$$f(x) = kx; \quad 0 \leq x \leq 4 \quad \text{Find (i) the value of } k; \text{ (ii) } P(x \geq 1); \text{ (iii) } P(x \leq 2)$$

MATHEMATICAL EXPECTATION

If X is a discrete or continuous random variable with probability function or probability density function $f(x)$, then the mathematical expectation of X is usually denoted by $E[X]$ or μ and defined by

$$\begin{aligned}\mu = E[X] &= \sum x f(x) ; \text{ If } X \text{ is a discrete random variable.} \\ &= \int_{-\infty}^{\infty} x f(x) dx ; \text{ If } X \text{ is a continuous random variable.}\end{aligned}$$

Properties of mathematical expectation of a random variable:

- (i) If b is a constant then $E[b] = b$
- (ii) If X is a random variable with expectation $E[X]$, then $E[aX+b] = a E[X] + b$, Where a and b constant.
- (iii) If X is a random variable with expectation $E[X]$, then $E[X-E(X)] = 0$
- (iv) If X and Y are random variables then $E[X+Y] = E[X] + E[Y]$
- (v) If X and Y are random variables then $E[X-Y] = E[X] - E[Y]$

Properties of variance of a random variable:

- (i) If b is a constant then $V[b] = 0$
- (ii) If X is a random variable, then $V[aX+b] = a^2 V[X]$, Where a and b constant.
- (iii) If X is a random variable with expectation $E[X]$, then
 $V(X) = E[X-E(X)]^2 = E(X^2) - [E(X)]^2$
- (iv) If X and Y are random variables then $V[X+Y] = V[X] + V[Y]$
- (v) If X and Y are random variables then $V[X-Y] = V[X] + V[Y]$

Application problem: Find mean variance and standard deviation of the following probability function:

Or, Find (i) $E[X]$ (ii) $V(X)$ and (iii) standard deviation

Values of $X:x$	0	1	2	3
$f(x)$	0.125	0.375	0.375	0.125

Solution: (i) **Mean:**

$$\begin{aligned}\text{We know, } E[X] &= \sum x f(x) \\ &= (0)(0.125) + (1)(0.375) + (2)(0.375) + (3)(0.125) \\ &= 1.5\end{aligned}$$

(ii) Variance:

$$\text{We know, } V(X) = E(X^2) - [E(X)]^2$$

$$\begin{aligned}\text{Here, } E(X^2) &= \sum x^2 f(x) \\ &= (0)^2(0.125) + (1)^2(0.375) + (2)^2(0.375) + (3)^2(0.125) \\ &= 3\end{aligned}$$

$$\begin{aligned}V(X) &= E(X^2) - [E(X)]^2 \\ &= 3 - (1.5)^2 = 0.75\end{aligned}$$

(iii) Standard deviation: $\sqrt{V(X)} = \sqrt{0.75} = 0.87$

Assignment problem: Suppose x is discrete random variable with probability function.

Values of X:x	-2	-1	0	1	2
f(x)	0.3	0.2	0.1	0.25	0.15

Compute (i) $E[x]$; (ii) $E[3x+3]$; (iii) $V[x]$ and (iv) $V[2x-3]$

Assignment problem: Let X be a continuous random variable with probability density function

$$f(x) = kx ; 0 \leq x \leq 4 \quad \text{Find (i) the value of } k; \text{ (ii) } P(x \geq 1); \text{ (iii) } P(x \leq 2)$$

(iv) Mean, variance and standard deviation of x